

Remittances Anchoring India's External Balances



Source:TH

Why in News?

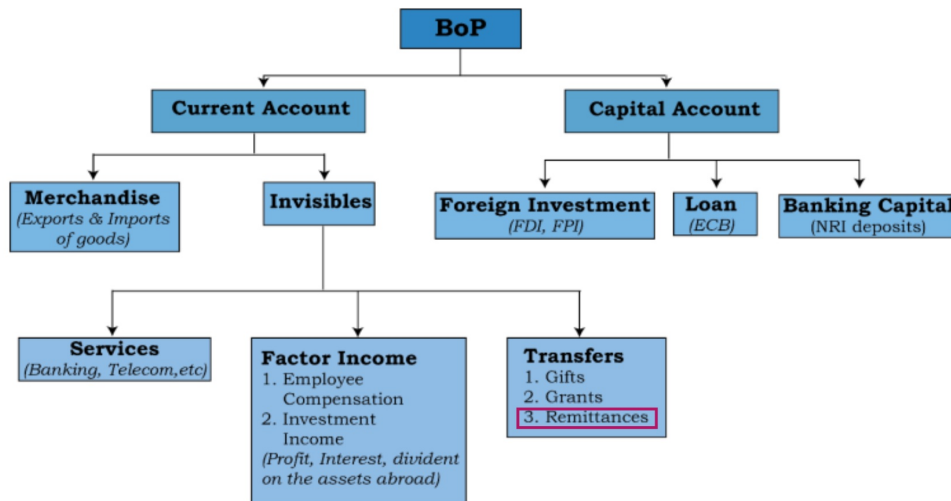
Economists have highlighted the critical role of **inward remittances** in managing **India's external balances**. Amidst global economic headwinds, volatile **Foreign Portfolio Investments (FPIs)**, and a widening merchandise trade deficit, **remittances have emerged as the unsung hero of India's macroeconomic stability**.

How do Remittances Anchor India's External Balances?

- **Cushioning the Current Account Deficit (CAD):**
 - **Balance of Payments (BoP) Structure:** The **BoP** is divided into the **Current Account and the Capital Account**. Remittances are categorized under the Current Account as "Invisibles."
 - In India's BoP, invisibles include services, income, and transfers. Remittances fall under private transfers/personal transfers, which are part of **Net Secondary Income (NSI)** in the Current Account.
 - **Offsetting Trade Deficit:** India is structurally a CAD economy driven by a **widening merchandise trade gap**, which ballooned from roughly USD 6 billion in 2000-01 to **USD 284 billion in 2024-25**. In 2024-25, remittances alone covered **47.5% of this massive trade deficit**.
 - Remittances act as a **massive, liability-free surplus** that offsets a significant portion of this trade deficit, keeping the overall **CAD within a manageable and safe threshold (usually targeted below 2.5% of GDP)**.
 - Since mid-2013, remittance inflows have, on average, financed more than the entire volume of India's merchandise trade deficit.
- **Stabilizing the Rupee:** Unlike FPIs, which are considered "**hot money**" and can trigger rapid capital outflows during global uncertainties (e.g., US Federal Reserve rate hikes), **remittances provide a stable and continuous supply of** foreign currency (like US Dollars, Euros, and UAE Dirhams) into the Indian economy.
 - When these foreign currencies are converted into INR for domestic use by families, it creates a persistent **demand for the Rupee**.
 - This buffers the domestic currency against sharp depreciations caused by volatile foreign investor sell-offs.

Significance for the Indian Economy

- **Non-Debt Creating Inflows:** Unlike **External Commercial Borrowings (ECBs)** or **Foreign Direct Investment (FDI)** which involve future repayment obligations or dividend repatriations, remittances are **private, non-returnable transfers**.
- **Forex Reserves:** They reduce the burden on the **Reserve Bank of India (RBI)** to actively intervene in the forex market, aiding in the accumulation and preservation of India's robust forex reserves.
- **Socio-Economic Development:** At the micro level, these **funds directly boost domestic consumption**, improve health and education outcomes for recipient families, and drive regional growth (historically notable in states like Kerala, Maharashtra, Karnataka, and Tamil Nadu).



What are Remittances?

- **Definition:** Remittances are cross-border financial transfers sent by migrant workers and **Non-Resident Indians (NRIs)** back to their home country to support families and local communities.
- **Driver:** These flows are largely driven by "**altruistic motives**" (**family obligations**) rather than "investment motives," making them highly stable and less vulnerable to sudden global financial shocks.
- **Regulatory Framework:** The **Foreign Exchange Management Act (FEMA), 1999** regulates all foreign exchange transactions in India.
 - Under the **Liberalized Remittance Scheme (LRS)**, a part of FEMA, Indian residents can remit up to **USD 250,000 per year for personal and investment purposes, with higher amounts requiring RBI approval**.
 - However, LRS prohibits remittances for **gambling, speculative trading, and terrorist financing**.
 - Remittances to India can be received through NRE (Non-Resident External), NRO (Non-Resident Ordinary), and **FCNR (Foreign Currency Non-Resident) accounts**.
- **Status of Remittances in India:**
 - **World's Top Recipient:** India has been the world's largest recipient of remittances since 2008.
 - **Massive Scale of Inflows:** Remittances reached **USD 135 billion in 2024-25**, comfortably outpacing gross FDI inflows, a trend that has held true for most years between 2000-01 and 2024-25.
 - **India's Remittance Profile:** According to the RBI survey on remittances published in 2025 (tracking data between 2016-17 and 2023-24), a decisive structural shift has occurred:
 - **Dominance of Advanced Economies:** Advanced nations (the US, UK, Singapore, Canada, and Australia) now account for more than 50% of total remittance inflows to India.

- **The Rise of the US:** The US has overtaken the UAE as the single largest source of remittances. The US share increased from 22.9% to **27.7%** .
- **Decline of the GCC Share:** The combined share of the Gulf countries has slipped down to **38%** , with the UAE's individual share falling from 26.9% to 19.2%.
- **Skill-Base Transformation:** This geographical shift indicates that high-wage, skilled professionals working in **OECD** countries are now out-remitting semi-skilled and unskilled blue-collar labor based in the Gulf.
- **Concerns:** The rising concentration of remittances from advanced economies increases India's vulnerability to host-country immigration policies and tightening visa regimes.
 - Furthermore, **rapid AI-driven automation in the West threatens** high-wage white-collar tech jobs that form the mainstay of recent remittance success.
 - Concurrently, geopolitical friction in the **GCC region** and local nationalization drives risk displacing large volumes of blue-collar workers.
 - Ultimately, any stagnation in these inflows amid **negative FDI and FPI trends will severely widen the current account deficit and destabilize the rupee.**

Frequently Asked Questions (FAQs)

1. What are remittances?

Remittances are cross-border money transfers sent by migrant workers and NRIs to their families or communities in India. They are usually private, non-returnable transfers.

2. Under which part of the Balance of Payments do remittances fall?

Remittances are part of the **Current Account** , specifically under **Net Secondary Income** or private transfers. They are not part of the Capital Account.

3. Why are remittances important for India's external sector?

They help offset the merchandise trade deficit, support the current account balance, and reduce pressure on the rupee and foreign exchange reserves.

4. How are remittances different from FDI and ECBs?

Unlike FDI or ECBs, remittances do not create repayment obligations, dividend outflows, or debt burdens. They are non-debt, creating inflows.

5. Which law governs remittances in India?

Foreign exchange transactions are regulated by **FEMA, 1999** . Under the **Liberalized Remittance Scheme (LRS)** , resident Indians can remit up to **USD 250,000 per year** for permitted purposes, subject to RBI rules.

[Watch Video on YouTube:

▶ <https://www.youtube.com/embed/W9DWraiu3IM>

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UPSC Civil Services Examination, Previous Year Questions

Prelims

Q. Which of the following constitute Capital Account? (2013)

1. Foreign Loans
2. Foreign Direct Investment
3. Private Remittances
4. Portfolio Investment

Select the correct answer using the codes given below:

- (a) 1, 2 and 3
- (b) 1, 2 and 4
- (c) 2, 3 and 4
- (d) 1, 3 and 4

Ans: (b)

Q. Which of the following is a most likely consequence of implementing the 'Unified Payments Interface (UPI)'? (2017)

- (a) Mobile wallets will not be necessary for online payments.
- (b) Digital currency will totally replace the physical currency in about two decades.
- (c) FDI inflows will drastically increase.
- (d) Direct transfer of subsidies to poor people will become very effective.

Ans: (a)

Mains

Q. 'Indian diaspora has a decisive role to play in the politics and economy of America and European Countries'. Comment with examples. (2020)

Summer Air Pollution in Indian Cities



Source: TH

Why in News?

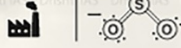
Summer is generally considered cleaner than winter, but in 2026 several Indian cities witnessed significant air pollution episodes. The reimposition of Stage-I restrictions (in May 2026) under **Graded Response Action Plan (GRAP)** in Delhi by the **Commission for Air Quality Management (CAQM)** highlighted the growing challenge of summer air pollution across urban India.

What are the Causes of Summer Air Pollution?

- **Regional Dust Storms (The Loo):** Intense solar heating over the Indian subcontinent creates a **vast low-pressure zone that stretches towards Iran and West Asia**.
 - **Impact:** The interaction between this low-pressure system and surrounding high-pressure zones generates **strong, dry, and scorching winds known locally as the loo**.
 - These powerful winds pick up massive quantities of desert dust from the **Thar Desert and the Arabian Peninsula**, transporting it thousands of kilometres across the **Indo-Gangetic plains** and drastically spiking **PM10** levels for days at a time.
- **Localized Thunderstorms (Andhi):** High convective heat causes sudden, short-lived local dust storms called **andhi**.
 - **Impact:** When strong, downward-moving cold air associated with summer thunderstorms hits the dry, moisture-depleted ground, it violently lifts loose topsoil and sweeps it through urban areas at high speeds.
 - While loo -type regional dust is dominant in North India, coastal and southern cities like Mumbai and Hyderabad frequently **encounter summer dust spikes due to these local thunderstorms**.
- **Accelerated Ground-Level Ozone (O₃) Formation:** **Ground-level ozone** is a secondary pollutant formed when **Nitrogen Oxides (NOx)** from **vehicles and Volatile Organic Compounds (VOCs)** from industries, paints, solvents, and fuel emissions react in the presence of strong sunlight.
 - **Heat Catalyst:** Intense summer sunlight and extreme heatwaves act as a **powerful thermal incubator** for this photochemical reaction.
 - As temperatures cross extreme thresholds, ozone levels spike dangerously during peak daylight hours.
- **Urban Heat Island (UHI) Effect:** The unchecked expansion of concrete surfaces, loss of green cover, and shrinking of **urban wetlands trap solar radiation within cities**.
 - This localized heating not only exacerbates ambient temperature spikes but also accelerates the atmospheric chemical reactions that form smog and ground-level ozone.
- **Resumption of Unregulated Construction Activities:** Severe emergency restrictions enforced during winter (such as the GRAP) are typically rolled back or eased as spring arrives.
 - **Consequence:** Once these curbs are lifted, heavy construction, infrastructure building, and demolition projects aggressively resume.
 - Without strict site-level controls like continuous water sprinkling, wind barriers, or dust sheets, these sites release massive amounts of coarse debris into the dry air.
- **Year-Round Anthropogenic Emissions:** While natural factors like wind and heat amplify summer pollution, the baseline fuel is provided by human activity.
 - Continuous vehicular exhaust, industrial chimneys, and the illegal burning of municipal solid waste or dry leaves in urban peripheries provide the steady stream of particulate matter and precursor gases needed to create toxic summer smog.

Air Pollutants

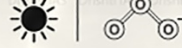
Sulphur Dioxide (SO₂)



It comes from the consumption of fossil fuels (oil, coal and natural gas). Reacts with water to form acid rain.

Impact: Causes respiratory problems.

Ozone (O₃)



Secondary pollutant formed from other pollutants (NO_x and VOC) under the action of the sun.

Impact: Irritation of the eye and respiratory mucous membranes, asthma attacks.

Nitrogen Dioxide (NO₂)



Emissions from road transport, industry and energy production sectors. Contributes to Ozone and PM formation.

Impact: Chronic lung disease.

Carbon Monoxide (CO)



It is a product of the incomplete combustion of carbon-containing compounds.

Impact: Fatigue, confusion, and dizziness due to inadequate oxygen delivery to the brain.

Ammonia (NH₃)



Produced by the metabolism of amino acids and other compounds which contain nitrogen.

Impact: Immediate burning of the eyes, nose, throat and respiratory tract and can result in blindness, lung damage.

Lead (Pb)



Released as a waste product from extraction of metals such as silver, platinum, and iron from their respective ores.

Impact: Anemia, weakness, and kidney and brain damage.

Particulate Matter (PM)



PM10: Inhalable particles, with diameters that are generally 10 micrometers and smaller.

PM2.5: Fine inhalable particles, with diameters that are generally 2.5 micrometers and smaller.

Source: Emitted from construction sites, unpaved roads, fields, fires.

Impact: Irregular heartbeat, aggravated asthma, decreased lung function.

Note: These major air pollutants are included in the Air quality index for which short-term National Ambient Air Quality Standards are prescribed.



How is Summer Pollution Different from Winter Pollution?

Feature	Summer Pollution	Winter Pollution
Primary Pollutants	Dominated by coarse particulate matter (PM10) and Ground-level Ozone	Dominated by fine particulate matter (PM2.5) and dense smog
Meteorological Drivers	Heatwaves, intense sunlight, dust storms, and high-velocity winds	Temperature inversion , stagnant cold air, low wind speeds, and fog
Major Sources	Wind-blown dust, construction activities, road dust, and vehicular emissions reacting with sunlight	Stubble burning , biomass burning for heating, industrial emissions, and trapped vehicular smoke
Pollution Formation Mechanism	Dust resuspension and photochemical formation of ozone under strong sunlight	Accumulation of pollutants due to poor dispersion and atmospheric trapping
Time of Peak Severity	Mid-day and hot afternoons (high ozone formation)	Early mornings and late nights (strong temperature inversion)
Key Seasonal Factor	Heat and sunlight-driven atmospheric chemistry	Cold weather and stagnant atmospheric conditions

What Can Cities Do to Combat Summer Air Pollution?

- **Dedicated Summer Action Plans:** Moving away from winter-only frameworks, cities should institutionalize 14-point and 8-point Summer Action Plans.
 - These plans focus on real-time tracking of industrial chimney emissions, anti-open-burning patrols, and strict enforcing of chemical guidelines for commercial paints and solvents to reduce ambient VOC leakages.
- **Leveraging Early Warning Systems:** While natural dust storms cannot be stopped, they can be predicted.
 - Tools like **Delhi's Air Quality Early Warning System (AQEWS)**, now extended to cities like Mumbai and Jaipur, provide multi-pollutant and weather forecasts several days in advance.
 - Municipalities must utilize these insights to issue timely public health advisories.
- **Active Construction Monitoring:** Stringent dust management protocols must be enforced year-round.
 - Utilizing digital infrastructure, such as the Brihanmumbai Municipal Corporation’s (BMC) **Air Quality Decision Support System (AQDSS)** , has enabled proactive tracking and enforcement actions against over 1,000 non-compliant sites since October 2025.
- **Targeting Ozone Precursors:** To reduce **ozone pollution** , cities must cut NOx and VOC emissions through cleaner transport, stricter industrial controls, and reduced vehicle idling.

Frequently Asked Questions (FAQs)

1. What is the main difference between summer and winter air pollution?

Summer pollution is mainly driven by **PM10 and ground-level ozone**, while winter pollution is dominated by **PM2.5 and dense smog** due to stagnant cold air.

2. Why does ozone increase in summer?

Ozone forms when **NOx** and **VOCs** react in the presence of strong sunlight and heat. Hot, sunny days accelerate this photochemical process.

3. What are the major causes of PM10 spikes in Indian cities during summer?

Major causes include **dust storms, road dust, construction activity, vehicular movement, and dry weather conditions** that resuspend loose particles.

4. What is the role of CAQM and GRAP in air pollution control?

CAQM implements and monitors air pollution control measures in NCR, while **GRAP** is a graded emergency response mechanism used to restrict pollution-causing activities.

[Watch Video on YouTube: [▶ https://www.youtube.com/embed/uyFaiYufBhM](https://www.youtube.com/embed/uyFaiYufBhM)]

UPSC Civil Services Examination, Previous Year Question (PYQ)

Prelims

Q. Artificial way of causing rainfall to reduce air pollution makes use of (2025)

- a) silver iodide and potassium iodide
- b) silver nitrate and potassium iodide
- c) silver iodide and potassium nitrate
- d) silver nitrate and potassium chloride

Ans: (a)

Q. How is the National Green Tribunal (NGT) different from the Central Pollution Control Board (CPCB)? (2018)

1. The NGT has been established by an Act whereas the CPCB has been created by an executive order of the Government.
2. The NGT provides environmental justice and helps reduce the burden of litigation in the higher courts whereas the CPCB promotes cleanliness of streams and wells, and aims to improve the quality of air in the country.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

Ans: (b)

Mains

Q. Describe the key points of the revised Global Air Quality Guidelines (AQGs) recently released by the World Health Organisation (WHO). How are these different from its last update in 2005? What changes in India's National Clean Air Programme are required to achieve revised standards? (2021)

India's 1st Blue Bond



Source: ET

Recently, **Sagarmala Finance Corporation** announced plans to issue India's first **blue bond** to diversify its funding sources and finance maritime and coastal infrastructure projects.

- **Blue Bond:** A blue bond is a **debt instrument** used to raise funds specifically for projects linked to **oceans, seas, coasts, rivers and other water-based ecosystems** .
 - It is generally targeted at investors interested in environmentally focused and sustainable finance instruments.
- **Purpose:** The funds will be used for lending towards the development of **maritime and coastal infrastructure** , including ports, port connectivity, shipbuilding, inland waterways and coastal road networks.
- **Blue Bond vs Green Bond:** A **green bond** is generally used to fund broader climate and environmental projects, while a **blue bond** specifically focuses on ocean, marine and water-related sustainable projects.
- **Global Examples:** Seychelles issued the **world's first sovereign blue bond in 2018** , while **Belize used a debt-for-ocean swap in 2021** to support marine conservation.
- **India's Relevance:** India's **long coastline, large fishing communities, port network and climate-vulnerable coastal regions** make blue bonds important for financing sustainable blue economy projects.
- **Blue Economy Link:** India's blue economy includes fisheries, shipping, offshore energy and coastal tourism, contributing about **4% to GDP and supporting around 4 million livelihoods** .
- **Significance:** Blue bonds can help India **attract global sustainable finance, reduce pressure on budgetary resources, support coastal communities** and balance maritime infrastructure growth with marine conservation.
- **Sagarmala Finance Corporation:** It is a **state-owned maritime-focused lender** established in **2016** under the **Ministry of Ports, Shipping and Waterways** . It received a **Non-Banking Financial Company (NBFC)** licence in **June 2025** .

Read More: [Blue Economy](#)

RudraM-II Missile Successfully Flight-Tested

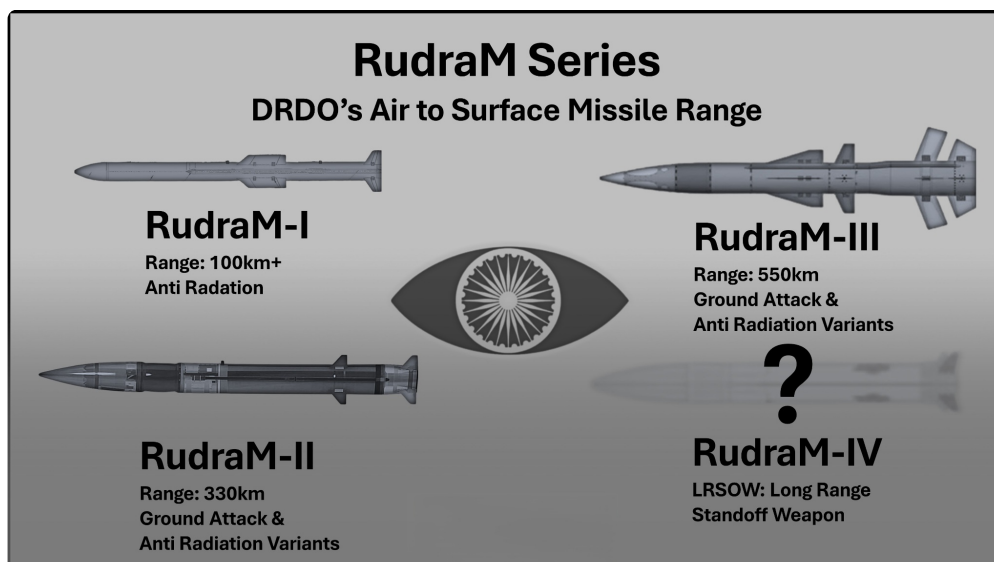


Recently, **DRDO** and the **Indian Air Force** successfully conducted flight-tests of the indigenous **RudraM-II air-to-surface missile** from an airborne platform, reportedly the **IAF's Su-30MKI** fighter aircraft.

- The missile was tested under extreme release conditions and met all mission objectives, with the trial supported by the **Integrated Test Range (ITR), Chandipur, Odisha**, a key DRDO facility for testing missiles, rockets and other defence systems.

RudraM-II Missile

- **About:** RudraM is a series of **indigenous air-to-surface anti-radiation missiles** developed to meet the Indian Air Force's requirement for enhancing its **Suppression of Enemy Air Defence (SEAD)** capability.
 - **Anti-radiation missiles** are designed to **target enemy radar, communication systems and other radio-frequency sources** that form part of hostile air-defence networks.
 - Three variants of the RudraM missile, **RudraM-I**, **RudraM-II** and **RudraM-III**, have been developed, with comprehensive testing of the first and second variants carried out earlier.
- **Purpose:** In SEAD operations, such missiles are used especially during the **initial phase of an air conflict** to strike enemy air-defence assets from stand-off ranges, improving the survivability of friendly aircraft in follow-on missions.
- **Development:** RudraM-II has been indigenously developed by **DRDO, with Research Centre Imarat (RCI), Hyderabad** as the nodal laboratory, in collaboration with several other DRDO establishments.
- **Significance:** The successful test strengthens the **IAF's precision strike capability**, enhances SEAD operations, supports Aatmanirbharta in defence, and reflects the growing maturity of India's advanced weapon technologies.



Read More: [Empowering India's Defence Sector](#)

